Optimization of Wind Turbine Airfoil Design Using XFOIL and openMDAO

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Collaborative Research: REU Site:
The Great Lakes Wind Energy Challenges

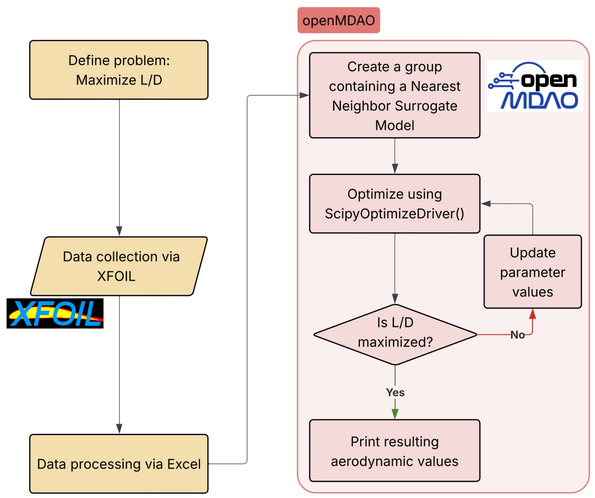
Abstract

This project aims to evaluate the benefits of using OpenMDAO and XFOIL to optimize wind turbine airfoil and blade design. Traditional optimization methods can become computationally intensive when incorporating parameters from multiple disciplines. OpenMDAO addresses this challenge by offering a modular, efficient framework for multidisciplinary design optimization. The project adopts a two-stage approach: (1) optimize airfoil geometry for maximum lift-to-drag ratio (L/D) using XFOIL and OpenMDAO, and (2) apply the optimized airfoil in a simplified blade model to estimate total lift under uniform inflow. Results show a maximum L/D of 74 and a total blade lift of 12,788 N, demonstrating OpenMDAO's potential to enable efficient, high-fidelity wind turbine design with reduced computational cost.

Methodology

Airfoil Optimization:

- Generated L/D data using XFOIL, varying max camber, max thickness, camber position, Reynolds Number (Re), and angle of attack (AOA) [1].
- Reduced *n_crit* from 9 to 5 in XFOIL to simulate more realistic, turbulent conditions and avoid idealized L/D values (~150).
- Used OpenMDAO's *SciPyOptimize Driver* with the *Sequential Least Squares Quadratic Programming (SLSQP)* algorithm for parameter optimization.
- Identified airfoil configuration that maximized L/D under the specified design space.



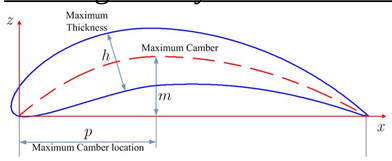
Blade Analysis:

- Assumed uniform freestream conditions: $V = 10\ m/s, p = 101.3\ kPa, and\ \rho = 1.225\ kg/m^3$ [1].
- Blade model with 30m radius, NACA 2411 airfoil, and constant pitch angle.
- Simplifications: Rotational effects, tip loss, and flow induction were not included.
- Total lift computed using trapezoid integration of:

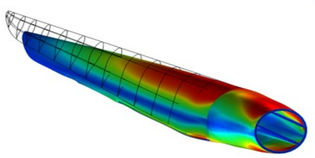
$$L_{tot} = \int_{r_{root}}^{R} \frac{1}{2}\rho V(r)^2 c(r) C_L(r) dr$$

- OpenMDAO components were developed to compute local relative velocity $V(r)$, chord distribution $c(r)$, and lift coefficient C_L [1].

Airfoil geometry

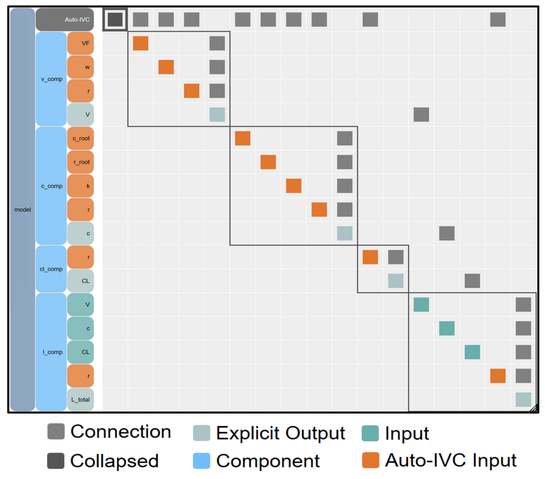


Wind turbine blade



Model Training

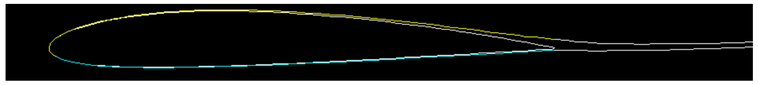
OpenMDAO Blade Analysis Model Structure (N2 diagram)

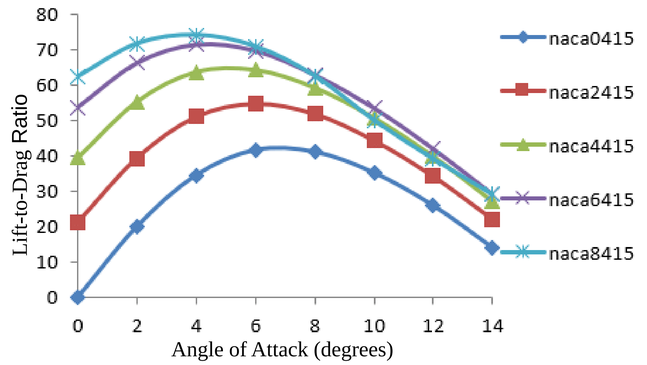


- Blade model used explicit components to compute variables and evaluate the total lift integral in a final aggregation component.
- Final component performed numerical integration for total lift.
- Airfoil model used a Weighted Nearest Neighbor (WNN) surrogate for fast, scalable interpolation [2].
- WNN predicts unknown values by averaging nearby known data points.

Results

- L/D optimization yielded NACA 2411 geometry, 8° AOA, and 1e6 Re as optimal [3].
- Max L/D = 74, with average runtime of 0.27 s.
- Blade analysis yielded a total lift = 12,788 N, runtime 0.09 s.
- Plot shows L/D vs. AOA for varying camber.





Discussions and Conclusions

Airfoil and Blade Analysis:

- L/D curve shows peak at 74 near 4° AOA, not at 8°, suggesting a mismatch between fitted model and raw data.
- Discrepancy likely due to sparse sampling, parameter coupling, or surrogate model interpolation.
- Results highlight the need for improved angle of attack resolution and Re-matching during training. Also, the induced flow and torque estimation are left for future work.

OpenMDAO:

- Supports flexible model expansion and variable testing.
- Enables fast model setup and execution.
- Well-suited for wind turbine design due to its multidisciplinary structure (aerodynamics, structure, cost, etc.).
- Offers a scalable, efficient framework for future wind energy optimization.

References

1. Leishman, J. G. (2023). Introduction to aerospace flight vehicles. Embry-Riddle Aeronautical University.
2. Welcome to OpenMDAO. OpenMDAO. (n.d.). https://openmdao.org/newdocs/versions/latest/main.html#
3. The relation curve of lift-to-drag ratio with camber. ResearchGate. (n.d.). https://www.researchgate.net/figure/The-relation-curve-of-lift-to-drag-ratio-with-camber_fig5_341023709